ITD 132-40 Structured Querry Language Semester Project

Project Overview

For your semester project, you will design and implement a complete database system demonstrating your mastery of SQL concepts and database design principles. You'll have the creative freedom to choose a domain that interests you, allowing you to build something meaningful while applying course concepts.

Why This Project Matters

This project simulates real-world database development scenarios you'll encounter in professional IT roles. Whether you're designing systems for e-commerce, healthcare, finance, or any other industry, the skills you develop here—database normalization, query optimization, and system documentation—are essential for any database developer, data analyst, or full-stack developer position.

Your project should showcase your ability to:

- Design and normalize a relational database
- Create and implement database tables with appropriate relationships
- Write complex SQL queries
- Apply database administration concepts
- Document and present your work professionally

Project Domain Selection

You have the freedom to choose any domain for your database project.

Some example domains include:

- Social Media Platform (like Instagram/TikTok/BeReal)
- Gaming Achievement & Stats Tracker (Steam/Xbox/PlayStation)
- Streaming Service (Netflix/Spotify/YouTube)
- Food Delivery App (DoorDash/UberEats)
- Fantasy Sports/Esports League
- Fitness & Workout Tracking App
- Anime/Manga Collection Database
- Digital Marketplace (like Etsy/Depop)
- Discord Server Management System
- Dating App Backend System

Domain Selection Guidance: *Choose a domain that is complex enough to require 8+ interconnected tables, but not so ambitious that it becomes unmanageable within a semester. Your domain should have clear entities, relationships, and realistic data requirements.*

Technical Requirements

Database Structure

- Minimum of 8 interconnected tables
- Proper implementation of primary and foreign keys
- Appropriate use of data types
- Implementation of constraints
- Indexes for performance optimization
- Must be in Third Normal Form (3NF) – eliminating data redundancy and ensuring each non-key attribute depends only on the primary key

Required SQL Components

- Table creation and modification statements
- Complex SELECT queries with joins
- Aggregate functions and GROUP BY clauses
- Subqueries and nested queries
- Views for common operations
- At least one stored procedure and at least one trigger
- Transaction management
- User permissions and security

Required Deliverables & Due Dates

1. Project Proposal & Domain Selection (Due: Week 5 - February 16th)

a. Detailed description of your chosen domain (minimum 250 words)
b. Initial list of 8-10 potential entities with descriptions
c. Preliminary relationships between entities identified
d. Project scope including: i. Types of users/roles ii. Key functionality to be implemented iii. Sample queries you plan to support
e. Rationale for domain selection f. Acknowledgment of academic integrity and AI usage policies
f. Submit as PDF through Canvas

2. Entity Relationship Diagram (Due: Week 8 - March 9th)

a. Professional ERD created using tools like: i. Lucid Charts (recommended) ii. Draw.io iii. Microsoft Visio iv. Other professional diagramming tools
b. Must include: i. All entities with attributes ii. Primary keys clearly marked iii. Foreign key relationships iv. Proper cardinality notation v. Attribute data types
c. Supporting documentation explaining: i. Design decisions ii. Normalization process iii. Any assumptions made
d. Submit both the diagram file and documentation PDF

3. Database Implementation (Due: Week 14 - April 20th)

a. Complete SQL script files including: i. Table creation statements with constraints ii. Index creation statements iii. Sample data insertion scripts (minimum 10 records per table) iv. Complex query examples demonstrating: - Multiple table joins - Subqueries - Aggregations - Views - At least one stored procedure and at least one trigger
b. Implementation documentation including: i. Table descriptions ii. Relationship explanations iii. Index justifications iv. Query descriptions
c. All code must be: i. Well-commented ii. Properly formatted iii. Free of syntax errors
d. Submit as both .sql files and documentation PDF

4. Final Presentation Package (Due: Week 16 - May 4th be with you)

a. 10-15 minute pre-recorded presentation using OBS or similar
b. Must include: i. Professional slide deck ii. Screen recording of database functionality demonstration iii. Live execution of key SQL queries iv. Clear audio narration v. Discussion of: - Design decisions - Challenges overcome - Lessons learned - Potential future enhancements
c. **Technical Documentation Package (to be submitted with presentation):** i. Complete ERD with any updates ii. All SQL scripts iii. Installation/setup guide iv. Sample queries with expected results v. User guide/manual vi. Documentation of AI tool usage (if applicable)

Academic Integrity & AI Usage

As outlined in the course syllabus, you may use AI tools to assist with your project, but you must:

- Disclose any AI tool usage in your documentation
- Use AI as a learning aid, not a replacement for your work
- Write original SQL code and database design
- Understand and be able to explain every aspect of your implementation

Remember that this project should demonstrate YOUR understanding of database concepts. While you can use AI tools for brainstorming or troubleshooting, the core design decisions and implementation should be your own work. Refer to the *"Artificial Intelligence Statement"* section of the course syllabus for complete details on permitted and prohibited AI usage.

Evaluation Criteria

Your project grade will be based on the following deliverables:

1. Project Proposal & Domain Selection (5%)

a. Clear project scope b. Feasible implementation plan c. Domain knowledge demonstrated d. Initial entity identification e. Thoughtful domain selection rationale

2. Entity Relationship Diagram (30%)

a. Completeness of entities and attributes b. Correct relationship mappings c. Proper cardinality notation d. Primary/foreign key identification e. Professional diagram presentation f. Appropriate normalization (3NF)

3. Database Implementation (40%)

a. SQL creation scripts

b. Sample data quality
c. Constraint implementation
d. Index usage
e. Complex queries demonstrating: Joins | Subqueries | Aggregations | Views | At least one stored procedure and at least one trigger

4. Final Presentation (25%)

a. Professional slide deck
b. Live demonstration
c. Presentation delivery

Important Notes

Each deliverable must be submitted by its respective due date to be eligible for full credit. Late submissions will be subject to the course late work policy as outlined in the syllabus.

Remember, this project represents 30% of your final grade and should demonstrate your cumulative learning from the course. Choose a project scope that challenges you while remaining manageable within the semester timeframe.

For questions or clarification about any aspect of the project, please contact your instructor through Canvas or email.